NARAYANA IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | SPCL_GTM-03 (FN) | Date: 13.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
B	C	C	C	D	B	A	B	D	B
11	12	13	14	15	16	17	18	19	20
C	C	A	A	A	A	C	A	C	C
21	22	23	24	25					
2	2	400	2	3					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	C	A	C	C	B	A	A	C	C
36	37	38	39	40	41	42	43	44	45
B	D	A	C	B	B	D	C	B	B
46	47	48	49	50					
2	51	0	100	2					

MATHEMATICS

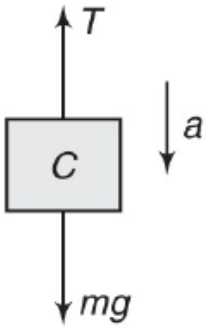
51	52	53	54	55	56	57	58	59	60
B	B	A	B	D	D	A	A	B	C
61	62	63	64	65	66	67	68	69	70
D	D	A	A	A	C	C	B	A	B
71	72	73	74	75					
2	2	32	2	4					

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PHYSICS

1. $$a=\frac{\text{Net pulling force}}{\text{Total mass}}$$

$$=\frac{(2+2-2)g}{2+2+2}=\frac{g}{3}$$



FBD of C

$$mg-T=ma=\frac{mg}{3}$$

$$\therefore \quad T=\frac{2}{3}mg$$

$$=\frac{2}{3}(20)=13.3N$$

2. Equating the volume, we have

$$8\left(\frac{4}{3}\pi r^3\right)=\frac{4}{3}\pi R^3 \text{ or } R=2r$$

$$v_T \propto (\text{radius})^2$$

Radius has become two times. Therefore, terminal velocity will become 4 times.

3. $$\frac{4}{5}mg-\frac{3}{5}qE=\frac{mv^2}{R}(i)$$

$$\frac{1}{5}mgR+\frac{3}{5}qER=\frac{1}{2}mv^2.....(ii)$$

$$\therefore \frac{mg}{qE}=\frac{9}{2}$$

4. $$v=\sqrt{\frac{\gamma RT}{M}} \propto \sqrt{\frac{T}{M}} \quad (\gamma=1.4 \text{ for both})$$

$$\therefore \quad \frac{T_{O_2}}{M_{O_2}}=\frac{T_{N_2}}{M_{N_2}}$$

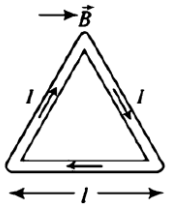
$$\therefore \quad T_{O_2}=\left(\frac{M_{O_2}}{M_{N_2}}\right)T_{N_2}$$

$$=\left(\frac{32}{28}\right)(273+15)$$

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$329K = 56^oC$

5. Since $\theta = 90^o$



Hence $\tau = NIAB = 1 \times I \times \left(\frac{\sqrt{3}}{4} l^2\right) B$

$= \frac{\sqrt{3}}{4} Il^2 B$

6. In term of energy considerations you can summarize the whole process as a loss in the gravitational potential energy of the block which is equal to work done against friction, of

$(mg) \times (3\sin 30^o) = (\mu mg \cos\theta) \times 2$

$\Rightarrow 3 \times \frac{1}{2} \Rightarrow \times \frac{\sqrt{3}}{2} \times 2 \Rightarrow \mu = \frac{\sqrt{3}}{2}$

7. $R \propto \frac{1}{r^2}$

$\frac{R_1}{R_2} = \left(\frac{r_2}{r_1}\right)^2$

$\frac{R_1}{R_2} = \frac{r_2^2}{r_1^2} or \frac{5}{R_2} = \frac{\left(3 \times 10^{-3}\right)^2}{\left(9 \times 10^{-3}\right)^2}; R_2 = 45\Omega$

These six wires are in parallel. Hence, the required resistance would b $R_2 = 7.5\Omega$

8. $n_1 = \frac{v}{2l_1}$

$l_1 = \frac{v}{2n_1}$

$n_2 = \frac{v}{2l_1}$

$l_2 = \frac{v}{2n_2}$

Now, $n = \frac{v}{2(l_1 + l_2)}$

Substituting the values we get, $n = \frac{n_1 n_2}{n_1 + n_2}$

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9. According to an ideal gas equation, the molecular weight of an ideal gas is

$$M = \frac{\rho RT}{P} \qquad \left(as\, P = \frac{\rho RT}{M} \right)$$

where P, T and ρ are the pressure, temperature and density of the gas respectively and R is the universal gas constant.

The molecular weight of A is

$M_A = \frac{\rho_A RT_A}{P_A}$ and thatof B is $M_B = \frac{\rho_B RT_B}{P_B}$

Hence, their corresponding ratio is $\frac{M_A}{M_B} = \left(\frac{\rho_A}{\rho_B}\right)\left(\frac{T_A}{T_B}\right)\left(\frac{P_B}{P_A}\right)$

Here, $\frac{\rho_A}{\rho_B} = 1.5 = \frac{3}{2}, \frac{T_A}{T_B} = 1\ and\ \frac{P_A}{P_B} = 2$

$$\therefore \frac{M_A}{M_B} = \left(\frac{3}{2}\right)(1)\left(\frac{1}{2}\right) = \frac{3}{4}$$

10. $v_e = \sqrt{2gR}$

$$\therefore \quad \frac{v_1}{v_2} = \sqrt{\frac{R_1}{R_2}} = \sqrt{\frac{R}{4R}}, \text{ or } \quad v_2 = 2v_1 = 2v_e$$

11. Radiation force = Momentum transferred per second by electromagnetic wave to the mirror.

$$= \frac{2I_{av}A}{c} = \frac{2\times(10)\times\left(20\times10^{-4}\right)}{\left(3\times10^{8}\right)} = 1.33\times10^{-10}\,N$$

12. From Brewster's law, $\mu = \tan i_p \Rightarrow \frac{c}{v} = \tan 60^o = \sqrt{3}$

13. In balancing condition, $\frac{R_1}{R_2} = \frac{l_1}{l_2} = \frac{l_1}{100 - l_1}$

$$\Rightarrow \frac{X}{Y} = \frac{20}{80} = \frac{1}{4} \text{ and } \frac{4X}{Y} = \frac{l}{100 - l}$$

$$\Rightarrow \frac{4}{4} = \frac{l}{100 - l} \Rightarrow l = 50\,cm$$

14. $T = \frac{t}{n}$ and $\Delta T = \frac{\Delta t}{n}$

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$$\therefore \quad \frac{\Delta T}{T}\times 100 = \frac{\Delta t}{t}\times 100$$

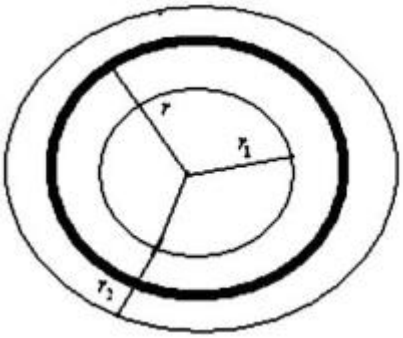
$$= \frac{0.1}{2\times 100}\times 100 = 0.05\%$$

$$= \frac{\Delta l}{l}\times 100 = \frac{0.1cm}{100cm}\times 100$$

Now, $T = 2\pi\sqrt{\frac{l}{g}}$ or $g = \frac{4\pi^2 l}{T^2} \propto \frac{l}{T^2}$

$\therefore$ % error in g = (% error in l) +2 (% error in T) = 0.1% + 2(0.05%) = 0.2 %

15.



$$H = \frac{-KAdT}{dr}$$

$$H = \frac{-K4\pi r^2}{dr}dT$$

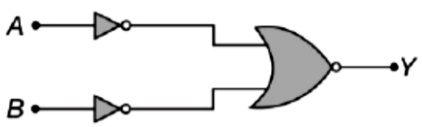
$$\frac{dr}{r^2} = \frac{-4\pi K}{H}dT$$

$$\int_{r_1}^{r_2}\frac{dr}{r^2} = \frac{-4\pi K}{H}\int_{T_1}^{T_2}dT$$

$$\Rightarrow\left[\frac{1}{r_1}-\frac{1}{r_2}\right] = \frac{-4\pi K}{H}\left[T_2 - T_1\right]$$

$$= H \propto \frac{r_1 r_2}{r_2 - r_1}$$

16.



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By de Morgan's theorem, $\left(\overline{\overline{A}+\overline{B}}\right)=A.B$

A	B	$\overline{A}$	$\overline{B}$	$\overline{A}+\overline{B}$	$\overline{\overline{A}+\overline{B}}$	A.B
1	1	0	0	0	1	1
0	0	1	1	1	0	0
0	1	1	0	1	0	0
1	0	0	1	1	0	0

This is the same as AND Gate of A and B.

17. AB process is isochoric

$\therefore Q_1 - nC_V\Delta T$

$=n\left(\frac{5}{2}R\right)(T_B-T_A)$

$=2.5\left(nRT_B-nRT_A\right)$

$=2.5\left(P_BV_B-P_AV_A\right)$

$=2.5P_0V_0$

BC process is isothermal

$\therefore Q_2=nRT_B\ln\left(\frac{P_i}{P_f}\right)$

$=P_BV_B\ln 2$

$=\left(2P_0V_0\right)(0.7)$

$=1.4P_0V_0$

Hence, $Q_{net}=Q_1+Q_2$

$=3.9P_0V_0$

18. Threshold voltage for a and b is same and it depends on frequency. So, $f_a=f_b$. Photo current of a and b are unequal and photo current depends an intensity. So, $I_a\neq I_b$

19. We have got, $P=\frac{1}{2}\rho Av^3t$

$\therefore\ P\propto v^3$

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20. If $\left|\vec{R}\right| = \left|\vec{a} + \vec{b}\right|$ then, $(a-b) \le R \le (a+b)$.

21. $B_y = \dfrac{\mu_0}{4\pi} \dfrac{2\pi/\left(2r^2\right)}{\left[(2r)^2 + (2d)^2\right]^{3/2}}, B_x = \dfrac{\mu_0}{4\pi} \dfrac{2\pi / r^2}{\left[r^2 + d^2\right]^{3/2}}$

Solve to get $\dfrac{B_x}{B_y} = 2$

22. Let $T_x =$ Tension in copper wire in x direction

$T_C \cos 30^\circ = T_S \cos 60^\circ$

$T_C \times \dfrac{\sqrt{3}}{2} = T_S \times \dfrac{1}{2}$

$\sqrt{3} T_C = T_S$ ..... (i)

In y direction

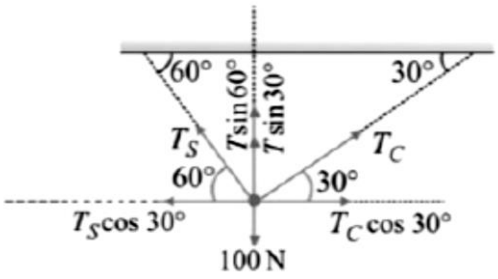
$T_C \sin 30^\circ + T_S \sin 60^\circ = 100$

$\dfrac{T_C}{2} + \dfrac{T_S\sqrt{3}}{2} = 100$ ..... (ii)

Solving Eqs. (i) and (ii)

$T_C = 50N$

$T_S = 50\sqrt{3}N$



We known $\Delta L = \dfrac{FL}{AY}$

$$\frac{\Delta L_C}{\Delta L_S} = \frac{T_C L_C}{A_C Y_C} \times \frac{A_S Y_S}{T_S L_S} = \left(\frac{T_C}{T_S}\right)\left(\frac{L_C}{L_S}\right)\left(\frac{Y_C}{Y_S}\right)$$

On solving above equation

$$\frac{\Delta L_C}{\Delta L_S} = 2$$

23. Flux varies with time and coefficient $\dfrac{4}{T}$ is common.

$$e = N\frac{d\phi}{dt} = N\frac{4}{T}\phi_m = 100 \times \frac{4}{1/50} \times 0.02 = 400V$$

24. $P = \dfrac{nE}{t} \Rightarrow 200 \times 10^6 = \dfrac{0.1}{100} \dfrac{mc^2}{24 \times 60 \times 60}$

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$\Rightarrow m = 192 \times 10^{-3}$ $\quad P = 2$

25. Comparing the given equation with

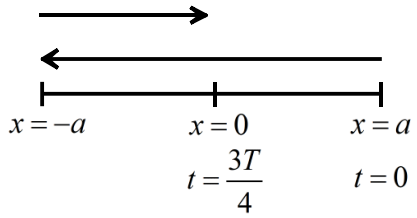
$$x = a\cos\omega t$$

$$\omega = \frac{\pi}{2}$$

or $\frac{2\pi}{T} = \frac{\pi}{2}$

$\therefore \quad T = 4s$

The given time $t = 3s$ is really $\frac{3T}{4}$

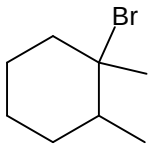


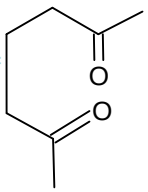
At time $t = 0$ particle is at $x = a$ (at extreme position) and at $t = 3s = \frac{3T}{4}$ it will be at mean position $x = 0$.

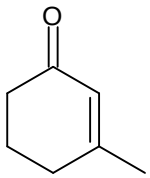
$\therefore$ Distance covered will be $3a$.

CHEMISTRY

26. Reactivity x magnitude of positive charge on diazonium salt

27. A = 

B = 

C = Aldol 

28. Ba, Ca belongs to IIA group, Se, S belongs to group VIA and as we go down the group ionisation enthalpy decreases with increasing atomic size SO, Ba<Ca and Se<S. Ar has stable completely filled electronic configuration and therefore has highest ionisation enthalpy.

Hence, the correct order is $Ba < Ca < Se < S < Ar$

In kJ/mol 503 590 941 999 1520

29. Reaction reagent used

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A) Hoffmann degradation $Br_2, NaOH$

B) Clemmensen reduction $Zn-Hg\ /\ HCl$

C) Cannizaro reaction Con. KOH, Δ

D) Reimer-tiemann reaction $CHCl_3, NaOH\ /\ H_3O^+$

30. $\left[Fe(CN)_6\right]^{4-}$ is diamagnetic due to d^6 strong field ligand where as $\left[Fe(CN)_6\right]^{3-}$ is paramagnetic due to presence of one unpaired e^-

31. Group 13 trivalent halides get easily hydrolysed by water due to their covalent nature.

32. Acidic strength of $SO_3 > SO_2$

No. of lone pair of e^- in O_2 molecule is 4

33. Total weight = 84+106=190

In 190 gm mixture weight of Na_2CO_3 = 106

19 gm of mixture of Na_2CO_3 = 10.6

34. $Na_2S_2O_3 + I_2 \longrightarrow 2NaI + Na_2S_4O_6$

$Cl_2 + {}^-OH \longrightarrow Cl^- + ClO_3^-$

35. All the molecules have $\mu_{net} \neq 0$

36. a) $\left[Co(C_2O_4)_3\right]^{-3}$: Co_{27}^{+3} : $3d^6 45^o$

Oxlate acts as string field ligand in this case

In this case

b) $\left[Fe(CN)_6\right]^{-3}$: Fe_{26}^{+3} : $3d^5$

${}^-CN$ is a strong field ligans

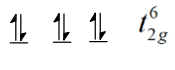
c) $\left[IrCl_6\right]^{-3}$: Ir^{+3} : $5d^6 65^o$

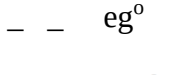
due to higher energy d-orbitals

we observe high CFS

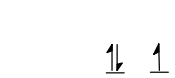
d) $\left[NiCl_4\right]^{-2}$: Ni^{+2} : $3d^8 45^o$

Cl^- is a weak field ligand









But in tetrahedral field we always have high spin complexes for $3d^8$ any way there can't be two cases.

37. Cr_{24} $[Ar]_{18}$ $3d^5$ $4s^1$

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$4s^1 \quad n=4 \quad l=0 \quad m=0 \quad s=1/2$

38. Acylation on benzene ring attached to activating group.

39. All are facts

40. Rate of the reaction depends on stability of Corbanion intermediate.

41. Resistivity $P=10\times10^{-3} ohm-cm$, molarity $M=0.4M$

$$\lambda M=\frac{1}{p}\times\frac{1000}{M}$$

$$=\frac{1}{10\times10^{-3}}\times\frac{1000}{0.4}=25\times10^4=25$$

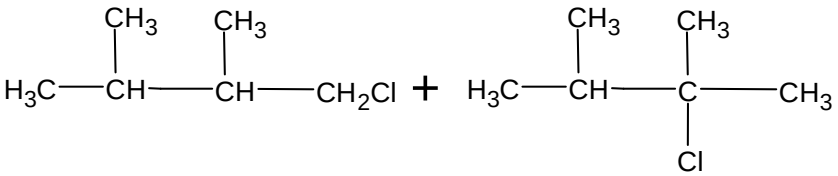
42. Balance reaction $10I^- + 2MnO_4^- + 16H^+ \longrightarrow 2Mn^{2+} + 8H_2O + 5I_2$

43. No. of alcoholic groups $=\frac{194-110}{42}=2$

44. $$K_p=\frac{(P_{co})^2(P_{O_2})}{(P_{CO_2})^2}$$

$$=\frac{\left(\frac{2}{6}.P\right)\left[\left(\frac{2}{6}\right)P\right]^2}{\left[\left(\frac{2}{6}P\right)\right]^2}=0.4$$

$$P=1.2 atm$$

46. 

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47.

A	B
1	1
$\downarrow$ 17 min	$\downarrow$ 51 min
$\frac{1}{2}$	$\frac{1}{2}$
$\downarrow$ 17 min	
$\frac{1}{4}$	
$\downarrow$ 17 min	
$\frac{1}{8}$	$\left(\frac{1}{8}\right)\times 4=\frac{1}{2}$

$\therefore$ After 51 min conc. Of B is 4 times that of A.

48. $\Delta S_{\text{surroundings}} = -nR\ell n\left(\frac{V_2}{V_1}\right) = -5\times 2\times \ell n\left(\frac{20}{10}\right)$

$$= -7\,cal\,.$$

$\therefore \quad x = -7$

$$x + 7 = 0$$

49. Range 33.30 – 33.50

$\frac{P_0 - P_s}{P_0}$ = constant on both containers

(Relative lowering of vapour pressure)

$$\frac{P_0 - P_s}{P_0} = X_{glucose}$$

$$\frac{P_0 - P_s}{P_s} = \frac{n_{glucose}}{n_{H_2O}} = \text{constant}$$

$\therefore$

$$\frac{\frac{10}{180}}{\frac{100-x}{18}} = \frac{\frac{20}{180}}{\frac{100+x}{18}}$$

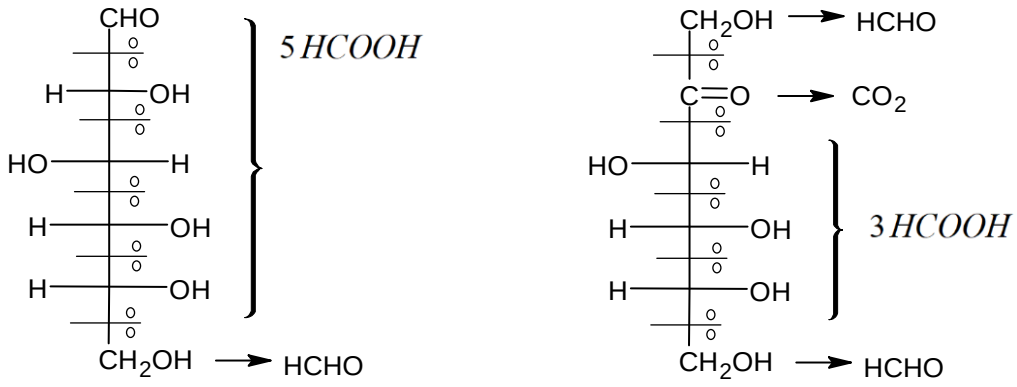
$$200 - 2x = 100 + x$$

$$3x = 100$$

$$x = \frac{100}{3} = 33.33\,g$$

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50.



glucose

Fructose

$5-3=2$

MATHS

51. As 1st difference of $T_1, T_2, T_3, \ldots$ are in AP

$T_1=3\ T_2=3+7=10,\ T_3=25$

$T_n=pn^2+qn+r$

$$\left.\begin{aligned} T_1&=p+q+r=3\\ T_2&=4p+2q+r=10\end{aligned}\right\}\rightarrow 3p+q=7$$

$$\left.T_3=9p+3q+r=25\right\}\rightarrow \underline{5p+q=15}$$

$2p=8\Rightarrow p=4\Rightarrow q=7-12=-5$ and $r=4$

$T_n=4n^2-5n+4$

$T_{20}=4(20)^2-5(20)+4=1504$

$T_{30}=4(30)^2-5(30)+4=3454$

$$\sum_{k=1}^{20}T_k=4\sum_{k=1}^{20}k^2-5\sum_{k=1}^{20}k+4\sum_{k=1}^{20}1$$

$$=\frac{4\times20\times21\times41}{6}-\frac{5\times20\times21}{2}+4\times20=10510$$

$$\sum_{k=1}^{30}T_k=4\sum_{k=1}^{30}k^2-5\sum_{k=1}^{30}k+4\sum_{k=1}^{30}1$$

$$=\frac{4\times30\times31\times61}{6}-\frac{5\times30\times31}{2}+4\times30=35615$$

52. Let $f(x)=(a+b-2c)x^2+(b+c-2a)x+(c+a-2b)$

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$a>b>c,\ \ a+b-2c>0$

Equation $f(x)=0$ has root in the internal (-1,0)

$\Rightarrow f(-1).f(0)<0$

$\Rightarrow (2a-b-c)(c+a-2b)<0$ .....(i)

Now, $2a-b-c=(a-b)+(a-c)$

So, from (i) , $c+a-2b<0\Rightarrow c+a<2b$ .....(ii)

Now, discriminant of equation $x^2+(a+c)x+4b^2=0$ is

$(a+c)^2-16b^2=(a+c)^2-4b^2-12b^2$

$=(a+c-2b)(a+c+2b)-12b^2<0 \quad (\because c+a<2b)$

53. For all $x\in\mathbb{R}$, we have

$f(2+x)=f(2-x)=f[7-(5+x)]$

$=f[7+(5+x)]=f(12+x)$

By replacing *x* with $x-2$ we get that $f(x)=f(x+10)$ for all $x\in\mathbb{R}$ .....(i)

Now, $0=f(0)=f(2-2)=f(2+2)=f(4)$ .....(ii)

From Equations (i) and (ii),we have $f(4+10n)=0$ for all integers *n*. Also, since $f(0)=0$, we have $f(10n)=0$ for all integers *n*. There are 403 integers of the form 10*n* and 402 integers of the form $10n+4$ in the interval $[-2010,2010]$. Therefore, there are atleast 805 integers *n* in $[-2010,2010]$ for which $f(n)=0$.

54. From the hypothesis we have $z=\frac{\sqrt{3}}{2}-\frac{i}{2}=i\left(-\frac{1}{2}-\frac{i\sqrt{3}}{2}\right)=iw$

Where $w=(-1/2)-(i\sqrt{3}/2)$ which is a cube root of unity.

Now, $z^{95}=(iw)^{95}=-iw^2$ (since $w^3=1$) and $i^{67}=i^3=-i$. Therefore,

$z^{95}+i^{65}=-i(1+w^2)=(-i)(-w)=iw$

$(z^{95}+i^{67})^{94}=(iw)^{94}=i^2w=-w$

Now, $-w=z^n=(iw)^n$

$\Rightarrow i^n.w^{n-1}=-1$

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$\Rightarrow n = 2,6,10,14,.....$ and $n-1 = 3,6,9,...$

Therefore $n = 10$ is the required least positive integer.

55. $$\int \frac{\sin\frac{7x}{2}}{\sin\frac{x}{2}}dx = \int \frac{2\sin\frac{7x}{2}\cos\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}dx$$

$$= \int\left(\frac{\sin 4x + \sin 3x}{\sin x}\right)dx$$

$$= \int \frac{\sin 4x}{\sin x}dx + \int \frac{\sin 3x}{\sin x}dx$$

$$= \int 4\cos x \cos 2x\,dx + \int (3 - 4\sin^2 x)dx$$

$$= \int 2(\cos 3x + \cos x)dx + \int (2\cos 2x + 1)dx$$

$$= \frac{2\sin 3x}{3} + 2\sin x + \sin 2x + x + c$$

56. It can be seen that $P^T = P^{-1}$

$$Q = PAP^T = PAP^{-1} \Rightarrow P^{-1}Q = AP^{-1}$$

Now, $$X = P^T Q^{2010} P = P^{-1}Q^{2010}P$$

$$= \left(P^{-1}Q\right)Q^{2009}P = A\left(P^{-1}Q^{2009}P\right)$$

$$= A\left(P^{-1}Q\right)Q^{2008}.P = A^2P^{-1}Q^{2008}P$$

Finally $$X = A^{2010}P^{-1}P = A^{2010}$$

Now $$A^2 = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$$

$$A^3 = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$$

By induction we can see that $A^n = \begin{bmatrix}1 & n\\ 0 & 1\end{bmatrix}$

Therefore $X = A^{2010} = \begin{bmatrix}1 & 2010\\ 0 & 1\end{bmatrix}$

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57. on solving first two equations $x=1, y=-1$ if equations are consistent $a-b-2=0$

58. $\because\ T_r(A)=a+b+c=10 \quad ; \quad (a\neq b\neq c)$

Therefore it may be (1,3,6), (1,4,5),(2,3,5)

$\therefore$ Number of matrices = $3!\times 3\times 3!$

Arrange diagonal elements arranging non-diagonal elements

$=3(3!)^2$

59. Term independent of x is ${}^nC_r\ a^{n-r}\ b^r,\ r=\dfrac{\frac{9}{6}}{\frac{1}{6}+\frac{1}{3}}=3$

60. Here, $f(x+y)=f(x)+f(y)$

$$f(2)=f(1)+f(1)=2$$

$$f(3)=f(1+2)=f(1)+f(2)=3$$

$$f(4)=f(1+3)=f(1)+f(3)=4$$

::

$f(x)=x$, for all $x\in R$

$\Rightarrow \quad f(\tan x)=\tan x,\ f(\sin x)=\sin x$

$$\therefore \quad \lim_{x\to 0}\frac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2 f(\sin x)}=\lim_{x\to 0}\frac{2^{\tan x}-2^{\sin x}}{x^2\sin x}$$

$$=\lim_{x\to 0}\frac{2^{\sin x}\cdot\left\{2^{\tan x-\sin x}-1\right\}}{\sin x.\,x^2}\times\frac{\tan x-\sin x}{\tan x-\sin x}$$

$$=\lim_{x\to 0}\frac{\left\{2^{\tan x-\sin x}-1\right\}}{\tan x-\sin x}\times\left\{\frac{\tan x-\sin x}{x^2\sin x}\right\}\times 2^{\sin x}$$

$$=\lim_{x\to 0}\frac{2^{\tan x-\sin x}-1}{\tan x-\sin x}\times\frac{1-\cos x}{x^2\cos x}\times 2^{\sin x}$$

$$=\lim_{x\to 0}\frac{2^{\tan x-\sin x}-1}{\tan x-\sin x}\times\frac{2\sin^2 x/2}{4(x/2)^2}\times\frac{2^{\sin x}}{\cos x}$$

$$=\log 2\times\frac{1}{2}\times 1=\frac{1}{2}\log 2$$

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61. Put $\pi x = t \Rightarrow dx = \dfrac{dt}{\pi}$

$$\therefore \quad I = \frac{1}{\pi}.\frac{\pi}{\pi}\int_0^{2008\pi} t|\sin t|\,dt = \frac{1}{\pi}\int_0^{2008\pi} t|\sin t|\,dt.....(i)$$

$$\Rightarrow \quad I = \frac{1}{\pi}\int_0^{2008\pi} (2008\pi - t)|\sin t|\,dt \quad .....(ii)$$

On adding equations (i) and (ii), we get

$$2I = \frac{2008\pi}{\pi}\int_0^{2008\pi} |\sin t|\,dt = (2008)^2.\int_0^{\pi} |\sin t|\,dt$$

$$\Rightarrow \quad I = (2008)^2 \qquad \Rightarrow \sqrt{I} = 2008$$

62. $y\sin 2x - \cos x + (1+\sin^2 x)\dfrac{dy}{dx} = 0$ where $y = f(x)$

$$\frac{dy}{dx} + \left(\frac{\sin 2x}{1+\sin^2 x}\right)y = \frac{\cos x}{1+\sin^2 x}$$

$$IF = e^{\int \frac{\sin 2x}{1+\sin^2 x}dx} = e^{\int \frac{dt}{t}} = e^{\ln(1+\sin^2 x)}$$

$$= 1+\sin^2 x \text{ (by putting } 1+\sin^2 x = t\text{)}$$

$$y(1+\sin^2 x) = \int \cos x\,dx$$

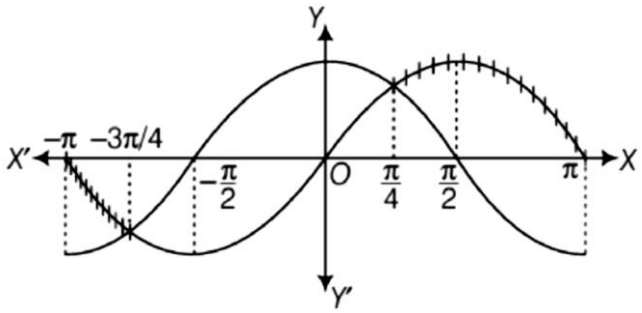
$$y(1+\sin^2 x) = \sin x + C;\ y(0) = (0) \Rightarrow C = 0$$

$$\therefore \quad y = \frac{\sin x}{1+\sin^2 x};\ y\left(\frac{\pi}{6}\right) = \frac{2}{5}$$

63. $$\frac{\text{Value of } (3x-2y+1) \text{ at } A}{\text{Value of } (3x-2y+1) \text{ at } B} > 0$$

$$\Rightarrow \frac{(\sin\alpha - 3) - (\cos\alpha - 2) + 1}{3-2+1} > 0$$

$$\Rightarrow \sin\alpha - \cos\alpha > 0 \Rightarrow \sin\alpha > \cos\alpha$$

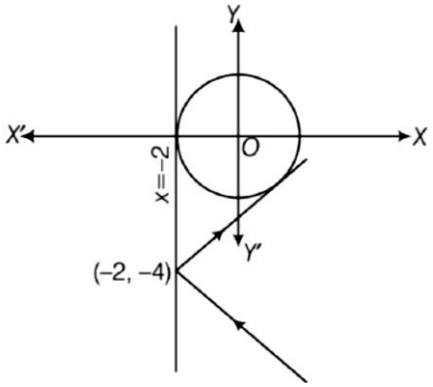


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It is clear from the figure $\alpha \in \left(-\pi, \frac{-3\pi}{4}\right) \cup \left(\frac{\pi}{4}, \pi\right]$

64. Any tangent of $x^2 + y^2 = 4$ is $y = mx \pm 2\sqrt{\left(1+m^2\right)}$

If it passes through $(-2,-4)$, then $-4 = -2m \pm 2\sqrt{\left(1+m^2\right)}$



or $(m-2)^2 = 1+m^2$

or $m = \infty,\ m = 3/4$

hence, the slope of the reflected ray is 3/4.

Thus, the equation of the incident ray is $y + 4 = -\frac{3}{4}(x+2)$

i.e., $4y + 3x + 22 = 0$

65. $\because$ $x - 2y + 4 = 0$ or $y = \frac{x}{2} + 2$

Condition of tangency $c^2 = a^2m^2 + b^2 \Rightarrow 4 = 4 \times \frac{1}{4} + b^2$

$\therefore$ $b = \sqrt{3}$

Now, common tangent meet on X-axis at a point $(-4,0)$. According to symmetry other common tangent meet Y-axis at $(0,-2)$.

$\therefore$ Equation of other common tangent is $\frac{x}{-4} + \frac{y}{-2} = 1$ or $x + 2y + 4 = 0$

66. As $0 \le \left(\sin^{-1} a\right)^2 \le \frac{\pi^2}{4}$, $0 \le \left(\cos^{-1} b\right)^2 \le \pi^2$,

$0 \le \left(\sec^{-1} c\right)^2 \le \pi^2 \left(\text{except } \frac{\pi^2}{4}\right)$ and $0 < \left(\text{cosec}^{-1} d\right)^2 \le \frac{\pi^2}{4}$

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So, $0<\left(\sin^{-1}a\right)^2+\left(\cos^{-1}b\right)^2+\left(\sec^{-1}c\right)^2+\left(\text{cosec}^{-1}d\right)^2\le\frac{5\pi^2}{2}$

$\therefore\left(\sin^{-1}a\right)^2+\left(\cos^{-1}b\right)^2+\left(\sec^{-1}c\right)^2+\left(\text{cosec}^{-1}d\right)^2=\frac{5\pi^2}{2}$ (Given)

$\Rightarrow\quad\left(\sin^{-1}a\right)^2=\frac{\pi^2}{4}$

$\left(\cos^{-1}b\right)^2=\pi^2$

$\left(\sec^{-1}c\right)^2=\pi^2$

and $\left(\text{cosec}^{-1}d\right)^2=\frac{\pi^2}{4}$

Hence, $\left(\sin^{-1}a\right)^2-\left(\cos^{-1}b\right)^2+\left(\sec^{-1}b\right)^2-\left(\text{cosec}^{-1}d\right)^2=0$

67. Combined mean $\bar{x}=\frac{n_1\bar{x}_1+n_2\bar{x}_2}{n_1+n_2}$

$=\frac{200\times25+300\times10}{500}=16$

Let $d_1=\bar{x}_1-\bar{x}=25-16=9$ and $d_2=\bar{x}_2-\bar{x}=10-16=-6$

Now we know that $\sigma^2=\frac{n_1\left(\sigma_1^2+d_1^2\right)+n_2\left(\sigma_2^2+d_2^2\right)}{n_1+n_2}$

$=\frac{200(9+81)+300(16+36)}{500}=\frac{33600}{500}=67.2$

68. Let (μ,μ,μ) be point of intersection of two lines.

$\Rightarrow\mu(\sin A+\sin B+\sin C)=2d^2$ .....(1)

and $\Rightarrow\mu(\sin2A+\sin2B+\sin2C)=d^2$ .....(2)

By $(2)/(1)$ $\sin A/2\sin B/2\sin C/2=\frac{1}{16}$

69. $P(\bar{S})=\frac{1}{3}$

$P(S)=2/3$

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$$P(S\cup R)=P(S)+p(R)-P(S\cap R)$$

$$P(R)=1/2$$

$$P(S)P(R)=1/3=P(S\cap R)$$

are Independent

70. $f'(0)=\sin^2(\sin 1);\qquad f''(0)=2\sin(\sin 1).\cos(\sin 1)\cos 1$

Now $g''(y)=\dfrac{-1}{\left[f'(x)\right]^3}f''(x)$

$$g''(3)=\frac{-f''(0)}{\left[f'(o)\right]^3}=\frac{-2\sin(\sin 1)\cos(\sin 1)\cos 1}{\sin^6(\sin 1)}$$

71. $\alpha.\hat{x}=\alpha.\beta-\alpha.\left(\alpha\times\vec{x}\right)=0$

Also, $\alpha\times\hat{x}=\alpha\times\beta-\alpha\times\left(\alpha\times\vec{x}\right)$

$$=\alpha\times\beta-\left(\alpha.x\right)\alpha+|\alpha|^2x$$

$\Rightarrow\beta-\vec{x}=\hat{\alpha}\times\beta+\vec{x}$ or $2x=\beta-\alpha\times\beta$

$$\Rightarrow\quad 4\left|x^2\right|=\left|\beta\right|^2+\left|\alpha\times\beta\right|^2-2\beta.\left(\alpha\times\beta\right)=2$$

$$\Rightarrow\quad |x|^2=\frac{1}{2}\qquad\Rightarrow 4|x|^2=2$$

72. Here, $y=\lim\limits_{n\to\infty}y_n=\lim\limits_{n\to\infty}\sum\limits_{m=1}^{n}\tan^{-1}\left(\dfrac{1}{2m^2}\right)$

$$=\lim_{n\to\infty}\sum_{m=1}^{n}\tan^{-1}\left(\frac{2}{1+(2m+1)(2m-1)}\right)$$

$$=\lim_{n\to\infty}\sum_{m=1}^{n}\tan^{-1}\left[\frac{(2m+1)-(2m-1)}{1+(2m+1)(2m-1)}\right]$$

$$=\lim_{n\to\infty}\sum_{m=1}^{n}\left(\tan^{-1}(2m+1)-\tan^{-1}(2m-1)\right)$$

$$=\lim_{n\to\infty}\left(\tan^{-1}3-\tan^{-1}1\right)+\left(\tan^{-1}5-\tan^{-1}3\right)+.....+\left(\tan^{-1}(2n+1)\right)-\tan^{-1}(2n-1)$$

$$=\lim_{n\to\infty}\left(\tan^{-1}(2n+1)-\tan^{-1}(1)\right)$$

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$$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Let $n \to \infty$, $P_n \to B$

$\because B \to \left(1, \frac{\pi}{4}\right)$, i.e.., coordinates of B approaches towards those of A.

$\therefore$ Slope of $AB = \left(\frac{d}{dx}\tan^{-1} x\right)_{\text{at } x=1}$

$$= \left(\frac{1}{1+x^2}\right)_{\text{at } x=1} = \frac{1}{2}$$

$\Rightarrow$ (Slope of AB)$^{-1}$ = 2

73.

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \qquad \& \qquad \frac{x^2}{9} - \frac{y^2}{16} = 1$$

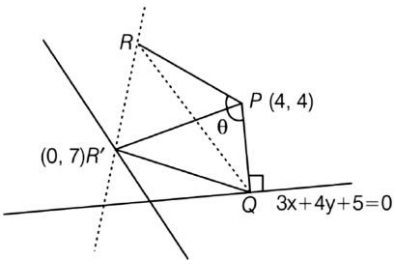
$$e_1 = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \qquad e_2 = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}$$

$$(e_1, e_2) = \left(\frac{\sqrt{7}}{4}, \frac{5}{3}\right) is\ a\ point\ on\ 16x^2 + 9y^2 = k$$

$$16x^2 + 9y^2 = 7 + 25 = 32$$

74. $f(x)$ is discontinuous in [1, 7] at two points i.e., x=5, 7

75. This will be maximum, if $\sin\theta = 1$ and *RP* is maximum



Since, line $y = mx + 7$ rotates about $(0,7)$, if PR' is perpendicular to the line than *PR*' is maximum value of *PR*.

$$\therefore \qquad m = -\left(\frac{4-0}{4-7}\right) = \frac{4}{3}$$

Hence, $3m = 4$

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